

Concept 2010 — BT's new generation dealing desk

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Picture a dealing room in any of the world's finance centres in which dealers conduct their business at desks that intelligently work 'with' them rather than merely 'for' them. Desks that give dealers and the finance institutions they represent a vital business edge. Desks that integrate the voice, data and video technologies and services that dealers require, for today and into the next millennium. Desks that manage the enormous flow of incoming information and present it to the dealers in a concise meaningful form. Desks that alert the dealers to sudden changes in the market and assist them in taking appropriate action. Desks that occupy the minimum of floor area and can be moved with minimum effort and cost. BT has developed and is now demonstrating such a desk. It is called Concept 2010.

1. Introduction

Today's wholesale finance sector relies heavily on state-of-the-art telecommunications to conduct business in an efficient and competitive manner. This is particularly evident in the trading community where deals are won or lost on the strength of the information that traders can access electronically across the globe, and the speed and effectiveness with which they can act on that information.

For many years BT has been a major player in this sector, and now commands a 55% share in the UK market for dealing room technology. With some 50 000 dealing positions installed at over 1400 sites in 45 countries, BT's market share world wide is 25%. In order to retain and indeed increase this coverage into the next millennium it is essential that BT enlightens its customers in the finance sector of the very latest in technologies and working practices. Furthermore, this enlightening process must be continuous and driven by the customers and the markets in which they operate. This can be achieved through the use of an evolving conceptual platform on which these technologies and working practices can be demonstrated and evaluated. BT Laboratories (BTL) [1] and Syntegra [2], the company's system integration arm, have jointly developed such a platform for the dealing desk market and it is called Concept 2010.

This paper describes the key features of Concept 2010, paying particular attention to those novel aspects that will bring real benefits to traders. Also presented is a vision of how this platform will evolve, with discussions of some of the new features currently under development.

2. The BT advantage

Everything about Concept 2010 is radical. The latest sound, video and data technologies have been integrated to enable traders to operate in the most flexible and productive manner. Flat panel displays are used throughout to make the most of screen real estate while at the same time reducing the overall size of the desk. Powerful data handling and visualisation techniques minimise the quantity of information presented to the dealer and maximise its content. Multipoint videoconferencing and a spatial sound system ensure that the dealer gets the most out of link-ups to remote locations. Speech platforms can be accommodated to allow the dealer to interact with and control the desk by spoken commands. The latest broadband wireless technologies do away with all cable communications links to the desk, thus eliminating the high cost incurred when moving desks and allowing the latest structured wiring techniques to be exploited. Finally, the 'furniture' aspect of the desk comprises the latest modular design which can be configured to any requirement.

Concept 2010 was unveiled at Innovation'97, a showcase exhibition held at BT Laboratories in April 1997. Since then the desk has been demonstrated in many of the world's dealing centres. Such has been the impact at these events that prospective customers now include some of the world's largest finance institutions.

3. Dealing platform

All dealing operations on the Concept 2010 desk are performed using the OTS (open trading system) market data platform, developed and marketed by Syntegra.

Such is the performance and flexibility of this platform that it has become an industry standard. OTS-4 is the latest version and is designed to work openly with all major market data systems. It gives the dealer unprecedented control over the presentation and manipulation of on-screen data, allowing him to use the display space to its best advantage (Fig 1). By incorporating a state-of-the-art object-oriented user interface, OTS-4 gives intuitive access to data and enables linking to other dealing applications in real time, thus ensuring that the dealer receives the most power from the desk and the best overall competitive advantage.

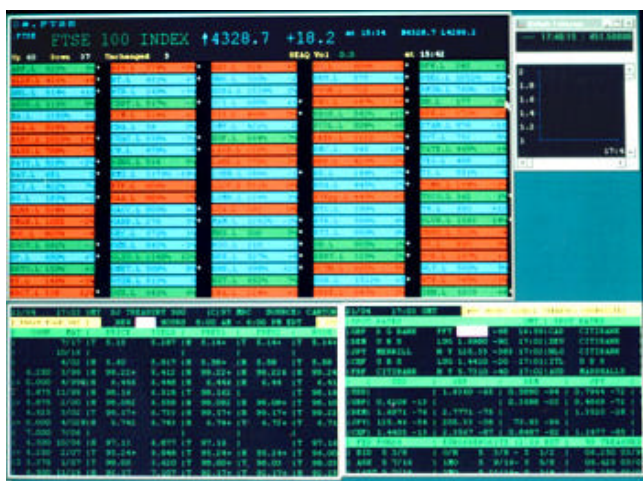


Fig 1 Typical OTS dealing data display.

The use of a voice/data integration (VDI) platform within the Concept 2010 desk means that a variety of time-critical functions can be performed automatically to assist the dealer. For example, when the dealer answers an incoming telephone call from a corporate client, that client's trading history, portfolio valuation and exposure will automatically appear on the displays. Similarly, if a telephone call is received from another dealer, the desk will automatically display the source of the call and the counter party information to make deal capture and settlement quicker and easier.

All of these features, including others described in later sections, are placed under the control of a Syntegra-developed Desk Manager. This provides the integration framework that 'glues' the various platforms together into a seamless whole, and provides dealers with an easy visual interface through which they can choose and manage the applications they require.

A crucial feature of the dealing platform within Concept 2010 is its compatibility and upgradability. As newer versions and different types of dealing software emerge they can be readily installed on the desk. For example, a platform currently under development within BT called INFER [3] offers many innovations that will benefit dealing and other finance operations. By ensuring that the desk can

incorporate developments as they emerge, the desk itself becomes future proofed and dealers are assured of a continued competitive advantage.

4. Displays

One of the most striking features of the Concept 2010 desk is its use of the latest flat panel displays. Gone is the need to accommodate deep, heavy, cathode ray tube (CRT) displays with their heat dissipation problems. Flat panel displays afford the minimum of depth, weight, heat and overall power consumption, while at the same time delivering a picture clarity and resolution that nowadays can easily match that of the best CRT displays. In addition, flat panel displays do not flicker as CRTs do and so afford comfortable viewing for much longer periods.

Concept 2010 utilises four 13.3 inch colour flat panel displays (Fig 2), manufactured by Hitachi and packaged to BT's specific requirements by Microvitec [4]. Each screen has an XGA resolution (1024 x 768 pixels), offering a picture real estate equivalent to that of a 17 inch CRT. By exploiting the latest super-TFT (thin film transistor) fabrication techniques these displays give a much brighter and sharper image than previous flat panel technologies, and a much wider viewing angle (140 ° horizontal and vertical). Three displays are mounted horizontally on the desk and effectively form a single panoramic viewing surface. The dealer may choose to display different information on each display, or instead may choose to 'spread' the same information across all three.

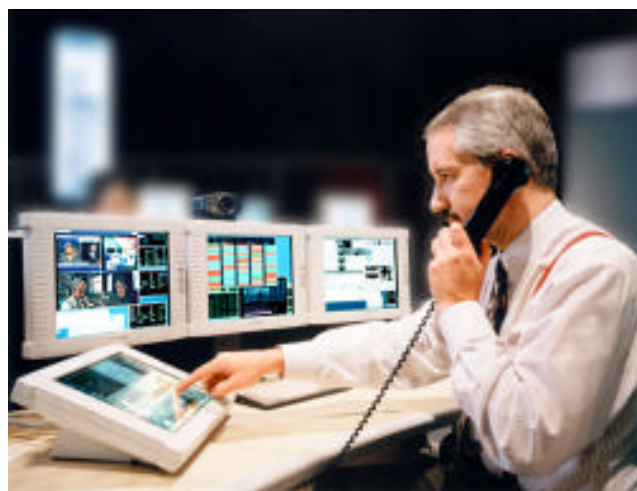


Fig 2 Flat panel displays and touch-screen.

The fourth display is mounted horizontally and is used as an active desk top. This display incorporates a touch-sensitive panel so that the dealer can set up and control many functions of the desk by touch. Fig 3 shows a typical layout that enables the dealer to control all telecommunications functions.



Fig 3 Typical touch-screen layout.

5. Videoconferencing

Modern day dealing is increasingly making use of videoconferencing and the Concept 2010 desk embraces this to the fullest extent. It can incorporate a multipoint videoconferencing system, such as the FreeVision system developed by C-C-C [5], which then enables the dealer to establish videoconference links with a number of colleagues simultaneously, and if necessary share their working data. Any of the vertical screens can be configured to display a videoconference image as an overlay, while still displaying dealing and market data in the background.

Fig 4 shows the C-C-C system in operation on a 4-way videoconference.



Fig 4 4-way videoconferencing.

By placing this system under the control of the Desk Manager described earlier, conference links can be set up manually by the dealer or automatically by the desk in response to, for example, a change in the dealer's position

which requires him to consult urgently with clients or colleagues.

In order to maximise the overall convenience of video-conferencing, the Concept 2010 desk uses a video camera that tracks the movement of the dealer. This ensures that the dealer is always ‘in shot’ from the point of view of the remote participants. This camera can be seen above the middle screen in Fig 2.

6. Data visualisation

Identifying market trends quickly and acting upon them in the correct manner is central to successful dealing, however, the volume of information that dealers are confronted with can be overwhelming. Concept 2010 utilises the latest in BTL-developed visualisation platforms to give the dealer complete control over how best to display data and market trends, and to highlight changes the instant they occur in a simple, intuitive manner.

One of the many visualisation options that Concept 2010 can offer dealers is ‘virtual worlds’ in which objects and surfaces can be made to represent either individual items of data or whole assemblages of data. The colour or ‘emotional behaviour’ of these objects and surfaces can be programmed to signify the state of the data they relate to. Fig 5 shows a typical example as viewed on one of the desk’s displays.

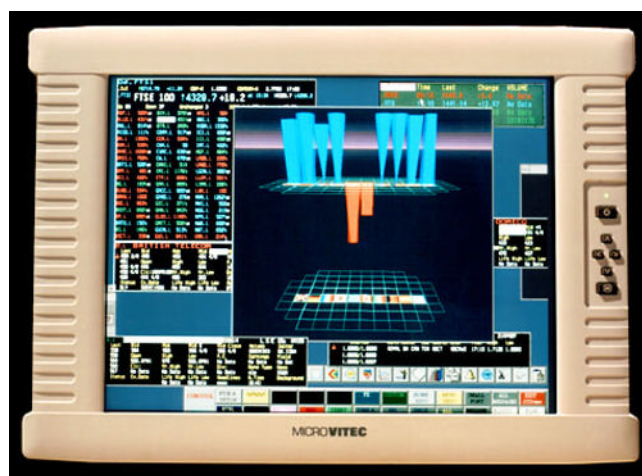


Fig 5 Data visualisation in a virtual world.

Individual stocks, bonds or currencies are represented in Fig 5 by vertical columns. The colour of each column signifies whether the value of that particular stock has risen (blue) or fallen (red) from its opening value, and the length defines the size of the shift in value. The shape of the columns indicates whether the value is rising (upward pointing triangle) at that precise moment or falling (downward pointing triangle). If the rate of this change in value exceeds a threshold (defined by the dealers themselves) these columns then also begin to wobble like

jelly, i.e. they look worried. This signifies to the dealers that urgent action is required.

Future developments in this field will see the use of 'intelligent agents' to perform search and retrieve functions and to build a profile of the dealer's pattern of usage so that the displayed information can be prioritised and formatted as preferred.

7. Spatial sound system

Voice telecommunications is a vital part of modern-day dealing, so much so that conventional dealing desks often incorporate an array of loudspeakers, each conveying a single audio link from, for example, an information supplier, a counter party, or fellow dealers at other finance centres, and so on. These loudspeakers unavoidably create clutter at the desk. The Concept 2010 desk removes this clutter by first mixing the various audio feeds and then delivering them to the dealer via a single pair of stereo loudspeakers. However, this on its own would have created new problems because all the simultaneous incoming voices would be effectively sitting on top of each other, thereby making it difficult or even impossible for the dealer to distinguish individual voices. To avoid this the desk incorporates a spatial sound system, developed specifically for this application by Edinburgh Communications Ltd.

This system accepts audio signals from up to 8 distinct sources and distributes them spatially around the desk in a 3-dimensional configuration chosen by the dealer (Fig 6). For example, the audio channel from New York could appear to emanate from the left of the desk, with London in the middle and Tokyo on the right, with other locations distributed in between. For dealing environments that require more than eight channels the system is readily scalable.

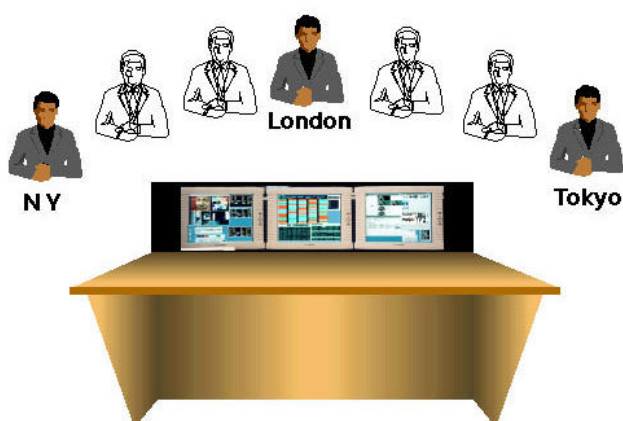


Fig 6 Spatial sound system.

The spatial sound system also improves the effectiveness of the videoconferencing facilities described earlier by creating the impression that the voices of the

remote participants emanate from the particular display screen on which they appear.

8. Furniture

The furniture constituents of the desk are the very latest in modular designs from Specialised Banking Furniture International Ltd (SBFI). This furniture can be configured exactly to the operating requirements of the dealers and their working environment.

The double-sided configuration shown in Fig 7 is particularly appropriate to Concept 2010 as it exploits the significant space-saving features of the flat panel displays. The distance between the vertical viewing surfaces on each side of the desk is only about 40 cm, which compares very favourably with the 150 cm or more for conventional desks that are populated by CRTs.

With Concept 2010 it is therefore possible to accommodate two dealing positions within a floor area that one conventional position would occupy. In a large dealing room this equates to an enormous saving in space and a reduction in operating costs.



Fig 7 Concept 2010 dealing desk.

A further important benefit afforded by the furniture is the unimpeded view that the dealers have of each other while seated at their desks. Since the flat panel displays have the same resolution and information area as the 17 inch CRTs used by many dealers today but are themselves only 13.3 inch, it is significantly easier to ensure that no equipment protrudes above the topmost element of the desk, thus ensuring full visual contact throughout the room.

9. Infra-red wireless communications

In dealing institutions it is vital that desks can be moved to different locations on the floor to respond to changes in the business requirements of dealing teams. Such moves might be frequent (e.g. monthly or even weekly), involving a great many desks and costing several thousand pounds per desk. The high cost is due in large part to the difficulty in recovering and relocating the under-floor communications cabling to the desk. Indeed, to minimise this cost the old cabling is often left in place and new cabling is overlaid to the new location. Clearly a point will be reached where the under-floor cabling cavities within the building are congested or floor loading becomes a concern, beyond which further moves are impossible without a major clear-out and re-cabling programme.

Concept 2010 addresses this problem by incorporating state-of-the-art wireless communications. The wireless channel is delivered to/from the desk via an infra-red communications system that is being developed jointly by BTL and Microvitec.

The 155 Mbit/s capacity of this system ensures that it is significantly future proofed and ATM compatible. Furthermore, the use of low-power lasers operating at a wavelength of 1550 nm ensures that the system has a good operating margin and is eye-safe at all times.

The infra-red wireless system creates an optical 'cell' around the desk from a transceiver located in the ceiling (Fig 8). In so doing, the system does away with the need for underfloor cabling and instead utilises the latest structured wiring techniques in the ceiling cavity. However, as the Concept 2010 desk is fully compatible with underfloor cabling, the dealing institutions can adopt the communications option that best suits their circumstances.

All voice, data and video to/from the desk are conveyed within the infra-red cell. The use of infra-red rather than radio as the wireless medium offers a unique evolutionary path because of the significantly greater bandwidth that is available. Infra-red also offers an element of security since the communications cell is confined to a single desk and the edge of the cell is very sharply defined.

With a sufficient number of ceiling stations distributed throughout the room (Fig 9), dealing desks can be moved to any other location without the need to recover and relocate cabling.

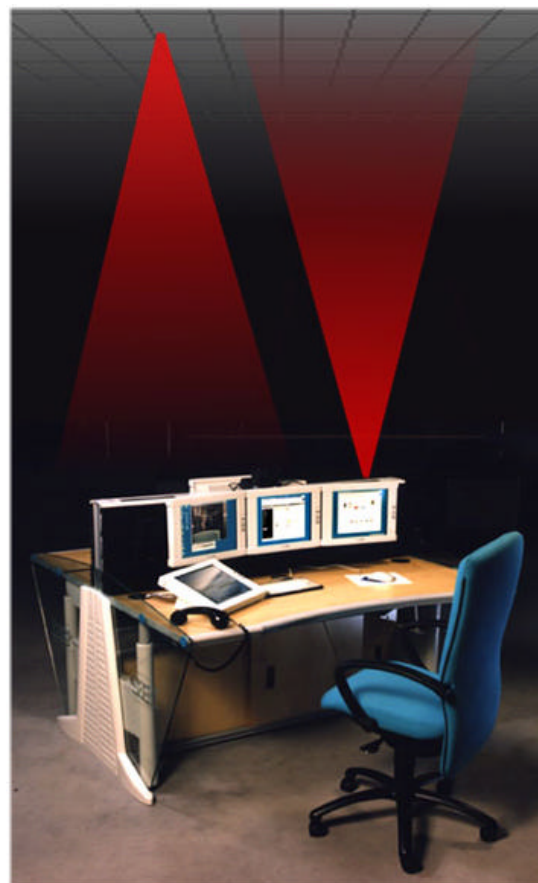


Fig 8 A representation of infra-red wireless communications to Concept 2010.

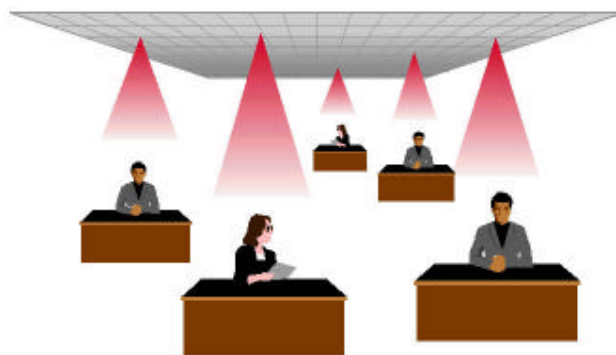


Fig 9 Infra-red wireless in a dealing room.

10. Modularity

A crucial feature of the Concept 2010 desk is modularity. All aspects of the desk, from the furniture through to the various technologies, can be configured to the exact requirements of individual customers. For example, the furniture can be free standing in a single-sided or double-sided form (see Fig 7 for the latter), or arranged end-to-end in rows or islands.

The mix of technologies at each dealing position is similarly flexible. The front position shown in Figs 2 and 7

makes full use of the latest technologies, with a vertical viewing surface that nearly spans the full width of the desk. In contrast, the rear position (Fig 10) uses a blend of new and present-day technologies. For example, the conventional dealer's console on the right of the desktop is changeable with the touch panel in use on the other position (Figs 2 and 7). The rear position also makes use of powerful data compression and visualisation techniques (Fig 5) to reduce the required vertical viewing surface to just one display. This configuration might therefore be particularly suited to dealing environments in which space is a premium and the width of the desks is less than that shown in Fig 7.



Fig 10 View of the rear dealer position.

11. Features under development

11.1 Speech recognition

At a conventional dealing desk the dealer will often have to look away from critical information on the displays in order to instigate a course of action. This may be as simple as a telephone call, or it could involve the setting up of a complex multi-way conference. In either case the dealer's attention is being distracted and hence their effectiveness is impaired to some extent. Speech recognition [6] offers a solution since it would allow dealers to speak commands to the desk without their attention being diverted.

The Concept 2010 desk can accommodate a variety of speech recognition systems, each with its own pros and cons in relation to the particular operating environment. These systems can be broadly categorised as speaker independent and speaker dependent. A speaker-independent system is generally the more robust of the two in the sense that it will respond to any speaker and will minimise the number of instances when a spoken command is incorrectly

interpreted. However, the vocabulary of such systems may be limited to a few hundred words and phrases. Conversely, a speaker-dependent system can accommodate a vocabulary spanning thousands of words and phrases, and also affords a degree of security since it will only respond to one dealer. However, misinterpretations and the need to repeat a spoken command might occur relatively more frequently.

Speech recognition could have a profound impact on dealing practice. For example, 'deal tickets' which carry all the details of a particular transaction could be completed in real time by the dealer largely or even entirely by voice command. Not only is this likely to speed up the process, but the data carried by the ticket could also include authentication of the dealer's voice in addition to the usual time and date stamps, at which desk the transaction was conducted, etc.

Dealers might also wish to use speech recognition to create documents by oral dictation. Such documents could, for example, give details on a transaction and could be sent to colleagues by e-mail for them to read at a convenient time.

11.2 Speech synthesis

When dealers are faced with visual information spread across multiple displays it is inevitable that critical events may occur on one of the displays that the dealers are not presently looking at. The conventional solution to this is for the desk to sound a simple audio prompt, such as a beep, and to overlay an accompanying message window on the relevant display. Although this approach is perfectly adequate, it represents another distraction of the kind alluded to in the previous section. Once again, speech can be used to remove this distraction but the roles are reversed as it is now the desk that speaks to the dealer by the use of speech synthesis.

The Concept 2010 desk can accommodate a variety of speech synthesis systems such as BTL's own Laureate system [6]. This system produces a very realistic voice which it then uses to speak commands and phrases from a library that is tailored to the particular application. The gender and timbre of the voice can also be selected to suit the application. With such a system integrated into the Concept 2010 desk, not only can all prompts to the dealer be vocal, but each prompt will state their exact nature, for example, 'the FTSE index has just risen by X points', or, 'you have an urgent call coming in from John in New York', etc. All of this represents a significant reduction in distractions and a corresponding increase in the effectiveness of the dealer.

Speech synthesis can overcome other distractions that commonly occur at dealing desks. For example, a

conventional desk will display streamed news from agencies such as Reuters as text on a display. To access this information the dealer must focus his attention on the relevant display. However, speech synthesis can be used to 'read aloud' streamed news to the dealer, which allows him to attend to other displays while he listens to the news.

11.3 Talking head

Speech synthesis systems such as Laureate can be linked to a virtual 'talking head' on one of the displays whose lip movements are synchronised to the synthetic words (Fig 11).



Fig 11 Virtual talking head.

This feature is being developed at BTL as a way of increasing the 'presence' of prompts and spoken information to the dealer. The head has the effect of bringing oral information to the foreground and so is a very powerful way of controlling the balance of importance between oral and visual information. For example, the desk might convey low importance prompts and audio messages in voice only, whereas critical prompts and messages might also be conveyed via the talking head. In parallel with this, the timbre of the voice could be selected to accentuate the level of urgency.

11.4 Text summariser

In addition to the 'streamed news' alluded to in the previous section, dealers may also wish to obtain information from various financial newspapers and journals. Nowadays this can be done with ease via Web browsers. However, whereas streamed news is highly abridged by the source agencies to maximise content within the fewest number of lines, news obtained directly from Web sites is largely unabridged. Consequently, dealers

spend significantly more time absorbing the latter forms of information, which in turn represents yet another distraction.

With applications such as this in mind, BTL has developed a text summariser called Netsum [7] which abridges streamed text or whole documents that have been downloaded. With text summation incorporated into the Concept 2010 desk, dealers can then receive abridged information from a wide range of news sources. Similarly, dealers could use text summation to reduce the length of a document that they wish to transmit to colleagues.

Text summarisers such as Netsum allow the extent and 'flavour' of the abridgement to be selected to suit the content of the original text. For example, a dealer might wish to abridge two multi-paged news articles, one concerning market trends, the other concerning a natural disaster somewhere in the world. Although both sets of information are important to the dealer because they may influence their subsequent decisions, their flavours are nevertheless very different. In order to produce abridgements that correctly encapsulate the overall meaning of the original texts different settings would be used.

Abridgements such as these can be delivered to the dealer as text on a display, or as a spoken message using the speech synthesis systems described earlier.

11.5 Security

It is vital that dealers and the institutions within which they operate have absolute confidence in the security of their dealing desks because of the extreme sensitivity of the data that they can access. This is particularly so if individual desks are time shared between several dealers. In the present realisation of the Concept 2010 desk security is effected by means of conventional user IDs and entry passwords when the dealer logs on. Although this level of security is entirely adequate new developments at BTL in the fields of iris recognition [8] and voice authentication [6] will raise this to an unprecedented level, thus maximising overall confidence. With these technologies integrated into the Concept 2010 desk a typical logging on might then comprise the following:

- the dealer sits at the desk,
- it scans his iris and recognises him to a very high certainty,
- the dealer then speaks a specific word or phrase,
- the desk recognises the voice and the specific password,

- the overall certainty that the dealer is bona fide is now absolute,
- the desk then activates and auto configures to the last known work pattern for that particular dealer.

A unique advantage that security systems like iris authentication and voice recognition have over conventional user IDs and passwords is that they can be in continuous operation. While the dealer is seated at the desk his voice is authenticated every time he speaks a command to the desk and his iris is periodically scanned. These features could be integral to a future voice-operated deal-capture system.

A further level of security might include active badges. During the logging on procedure the desk interrogates the dealer's badge via radio or infra-red, depending on the particular badge technology in use. Such badges could also be part of a building-wide security and tracking system.

12. Conclusions

This paper has shown that Concept 2010 represents a truly radical approach to dealing desk design. The use throughout of state-of-the-art technologies and the latest in software platforms ensures that dealers have the best possible working environment and competitive edge. Furthermore, the modular design of the desk allows the mix of technologies, services and furniture to be tailored to the exact requirements of individual dealers and institutions. Indeed, this modularity also means that the desk can be configured for a wide range of other key applications, for example, command and control centres, multimedia work stations, etc.

As the majority of the desk features described in this paper are available today as off-the-shelf items, dealers whose requirements are satisfied by that particular technology mix can be accommodated today. The other features will be introduced when the underlying technologies become sufficiently robust. Indeed, Concept 2010 is a demonstration platform that will continue to evolve. For example, the use of virtual worlds in dealing is a new paradigm being pioneered by BT. The value it adds to a dealer's effectiveness is still being evaluated, but as the 'Sega' generation start to appear on the dealing room floor, the benefits will become more quickly realised. Another example is the use of speech platforms. The dealing environment is arguably one of the most challenging for speech applications, which means that Concept 2010 is the ideal platform to demonstrate and evaluate these new ideas.

Through the facilities afforded by Concept 2010, BT now has a powerful demonstration and evaluation platform designed to show strategic customers in the finance sector the latest technologies and applications in a relevant environment. The feedback and validation that this will

generate will help BT to take those customers into the next millennium, and continue to offer solutions that give an assured competitive edge.

Acknowledgements

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David Heatley obtained BSc and MSc degrees in 1978 and 1981 respectively, and a PhD in Optical Communications Systems in 1989. He joined BT Laboratories in 1978 to work on the development of analogue and digital optical fibre systems for video and broadband services. In 1985 he was appointed to head a group responsible for the development of optical receivers for terrestrial and undersea fibre systems. In this capacity he was a member of the team that received the Queen's Award for Technology in 1990. He is presently with the Networks and Technology Centre and heads a group

with special responsibility for mobile telecommunications systems and future technologies. During his career he has published widely on telecommunications, and aspects of this work now feature in a recently released book he co-edited on future telecommunications systems. He is a Member of the IEE and a Chartered Engineer.



Ian Cockburn joined BT in 1992, prior to which he had spent 22 years in the industry moving from engineering through programming to project management. While with BT has been involved in many bids and projects. Most recently he has been associated with Syntegra's OTS Trading Systems product for the Wholesale Finance sector. He is currently the Syntegra project manager for Concept 2010 and sits on the steering committee for the Advanced Development Programme. He is a member of the APM.



Terry Lyne joined BT laboratories (BTL) in 1979 after obtaining a BSc in Electronic Engineering from Birmingham Polytechnic. He has worked in a number of component and test areas including the design of automated test equipment and the management of a semiconductor test facility.

He has worked on AI systems to support BT's field technicians and he is the holder of British and European Patents for telephone switching systems. His current work centres on the Concept 2010 project where he has special responsibility for co-ordinating much

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Throughout his career he has been an active member of various engineering Community support organisations and he is currently on the National panel of Judges for Young Engineers for Britain.



Kim Fisher qualified as a product designer in 1975, with a BA honours degree in Industrial Design Engineering. He then worked for Russell Hobbs moving to Gillette UK Research Laboratories in 1978 and then Pye Telecom in 1979 (later Philips Radio Communication Systems).

In 1985, he joined BT Laboratories to set up the industrial design unit, becoming industrial manager for the BT Products Division in 1989.

In April 1990, he joined the new advanced concepts unit in the Systems Research division and is currently working in the Centre for Human Communications at BT Laboratories.



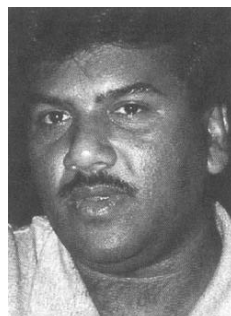
Ian Neild graduated from Hull University and started work for BT in 1993.

He initially worked on Gbit wireless links and is currently working on WLAN and IrDA data solutions for use in payphones and as a precursor to UMTS.



Alex Williamson joined BT in 1978 as an apprentice and then took up full time duties as a field maintenance engineer. Following this he became an instructor and lecturer at BT International in a variety of transmission subjects.

For some years now he has been with National Business Communications providing technical sales support for front office trading room systems. As part of his present duties he provides specialist input to the Concept 2010 project.



Nasir Haque obtained a BSc in Computer Systems Engineering in 1986, an MSc in Microelectronics System Design in 1988, and an MBA in 1991.

While with BT he has been involved in many developments which have been down-streamed as successful products. He is currently a Trading System Consultant for System Integration Sales.